Experiment No. 3

Test Case 1

Design a quarter-wave transformer for an input impedance (Zin) of 50 ohms and a load impedance (ZL) of 100 ohms. Analyze the transformer's performance as the operating frequency is varied across {100 MHz, 200 MHz, 300 MHz, 500 MHz}.

PROGRAM:

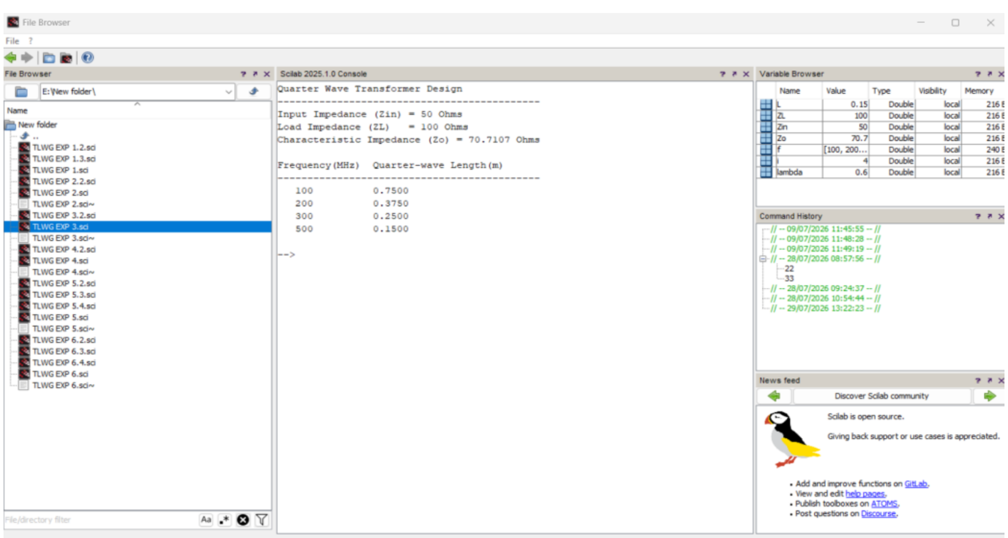
```
clear; clc;
ZL=100; Zin=50; f=[100,200,300,500]*10^6;
Ro=sqrt(Zin*ZL);
printf("characteristic impedance Ro = %f ohms\n",Ro);
for i=1:4
  L=300/(f(i)*10^-6)/4;
  printf("f=%f MHz  length = %f m\n",f(i)*10^-6,L);
end
```

OUTPUT:

ZL = 100 ohms, Zin = 50 ohms → Characteristic impedance Ro = 70.7107 ohms (Simulation) / 70.7107 ohms (Theoretical)

Frequency (MHz)	Length of λ/4 transformer (m) — Simulation	Length of λ/4 transformer (m) — Theoretical
100	0.7500	0.7500
200	0.3750	0.3750
300	0.2500	0.2500
500	0.1500	0.1500

OUTPUT SCREENSHOT:



Test Case 2

For a fixed operating frequency of 2 GHz and a load impedance (ZL) of 100 ohms, design a quarter-wave transformer and evaluate its performance as the input impedance (Zin) varies through {25 ohms, 50 ohms, 75 ohms, 100 ohms}. Validate that the transformer's characteristic impedance (Ro = √(Zin·ZL)) changes with Zin.

PROGRAM:

```
clear; clc;
ZL=100; f=2*10^9; Zin=[25,50,75,100];
for i=1:4
  Ro=sqrt(Zin(i)*ZL);
  L=300/(f*10^-6)/4;
  printf("Zin=%f  Ro=%f  Length=%f m\n",Zin(i),Ro,L);
end
```

OUTPUT:

ZL = 100 ohms, f = 2.0 GHz — Variable: Zin

Zin (Ω)	Ro (Ω) — Simulation	Length (m) — Simulation	Ro (Ω) — Theoretical	Length (m) — Theoretical
25	50.0000	0.0375	50.0000	0.0375
50	70.7107	0.0375	70.7107	0.0375
75	86.6025	0.0375	86.6025	0.0375
100	100.0000	0.0375	100.0000	0.0375

OUTPUT SCREENSHOT: